1 I

1.1 (1)

Since the Hamiltonian is separable in the three spatial directions, the single-particle partition function can be written as:

$$\begin{aligned} z_1 &= \frac{1}{h^3}\int d^3p\, d^3r\, \exp\left[-\beta\left(\frac{\boldsymbol{p}^2}{2m}+\frac{m\omega^2\boldsymbol{r}^2}{2}\right)\right] \\ &= \left[\frac{1}{h}\sqrt{\frac{2\pi m}{\beta}}\sqrt{\frac{2\pi}{\beta m\omega^2}}\right]^3 \\ &= \boxed{\left(\frac{1}{\beta\hbar\omega}\right)^3}. \end{aligned}$$

1.2 (2)

The three quantum numbers are non-negative integers:

$$n_x, n_y, n_z = 0, 1, 2, \ldots.$$

2 II

2.1 (3)

From the condition $U(l_{\text{th}}/\sqrt{\pi}) = k_BT$, we have:

$$\frac{1}{2}m\omega^2\frac{l_{\text{th}}^2}{\pi} = k_BT.$$

Thus:

$$\boxed{l_{\text{th}} = \frac{1}{\omega}\sqrt{\frac{2\pi k_BT}{m}}}.$$

Similarly, using the condition that the kinetic energy associated with $|\boldsymbol{p}| = h/(\sqrt{\pi}\lambda_{\text{th}})$ equals k_BT, we obtain:

$$\frac{h^2}{2\pi m\lambda_{\text{th}}^2} = k_BT.$$

Therefore:

$$\boxed{\lambda_{\text{th}} = \frac{h}{\sqrt{2\pi m k_BT}}}.$$

2.2 (4)

Using the results of (3), we obtain:

$$\begin{aligned} \frac{\lambda_{\text{th}}}{l_{\text{th}}} &= \frac{h\omega}{2\pi k_BT} \\ &= \frac{\hbar\omega}{k_BT} = \boxed{\beta\hbar\omega}. \end{aligned}$$

3 III

3.1 (5)

For N indistinguishable, non-interacting particles in classical statistical mechanics, the partition function is:

$$Z = \frac{z_1^N}{N!} = \frac{1}{N!}\left(\frac{1}{\beta\hbar\omega}\right)^{3N}.$$

The Helmholtz free energy is:

$$\begin{aligned} F &= -k_B T \ln Z \\ &= k_B T \left[\ln N! + 3N \ln(\beta\hbar\omega)\right] \\ &\approx N k_B T \left[\ln N - 1 + 3\ln(\beta\hbar\omega)\right], \end{aligned}$$

where we have used Stirling's approximation $\ln N! \approx N \ln N - N$.

Therefore:

$$\begin{aligned} -\frac{\beta F}{N} &\approx 1 - \ln\left[N(\beta\hbar\omega)^3\right] \\ &= \boxed{1 - \ln(\rho\lambda_{\text{th}}^3)}. \end{aligned}$$

3.2 (6)

The expected particle number can be separated into the ground-state and excited-state contributions:

$$\langle N\rangle = \frac{1}{e^{\beta(e_g-\mu)}-1} + \sum_{\text{excited states}} \frac{1}{e^{\beta(e-\mu)}-1}.$$

In the continuous-energy approximation, the sum over excited states becomes an integral over the density of states:

$$\begin{aligned} \langle N\rangle &= \frac{1}{e^{\beta(e_g-\mu)}-1} + \int_{e_g}^{\infty} \frac{D(e)}{e^{\beta(e-\mu)}-1}\, de \\ &= \frac{1}{e^{\beta(e_g-\mu)}-1} + \frac{C}{(\hbar\omega)^3}\int_{e_g}^{\infty} \frac{(e-e_g)^2}{e^{\beta(e-\mu)}-1}\, de. \end{aligned}$$

Let $x = \beta(e - e_g)$ and $A = \beta(e_g - \mu) > 0$. Then:

$$\begin{aligned} \langle N\rangle &= \frac{1}{e^{A}-1} + \frac{C}{(\beta\hbar\omega)^3}\int_{0}^{\infty} \frac{x^2}{e^{x+A}-1}\, dx \\ &= \frac{1}{e^{A}-1} + \frac{C}{(\beta\hbar\omega)^3} I(A), \end{aligned}$$

where $I(A)$ is the function defined in the problem.

Using $\lambda_{\text{th}}/l_{\text{th}} = \beta\hbar\omega$, we finally obtain:

$$\boxed{\rho\lambda_{\text{th}}^3 = \frac{(\beta\hbar\omega)^3}{e^{A}-1} + C I(A), \qquad A = \beta(e_g - \mu).}$$

3.3 (7)

From the definition of the density parameter, we have:

$$\rho\lambda_{\mathrm{th}}^3 = \langle N\rangle \left(\frac{\lambda_{\mathrm{th}}}{l_{\mathrm{th}}}\right)^3 = \langle N\rangle(\beta\hbar\omega)^3.$$

When $\rho\lambda_{\mathrm{th}}^3$ is fixed and $\langle N\rangle \gg 1$, we obtain:

$$(\beta\hbar\omega)^3 = \frac{\rho\lambda_{\mathrm{th}}^3}{\langle N\rangle} \ll 1.$$

Thus:

$$\boxed{\beta\hbar\omega \ll 1}.$$

Next, consider the ground-state population:

$$N_0 = \frac{1}{e^A - 1}, \qquad A = \beta(e_g - \mu) > 0.$$

Under the stated assumption that the separation $e_g - \mu$ does not become vanishingly small in the macroscopic limit (with A remaining finite and positive), $N_0 = O(1)$. Hence:

$$\frac{N_0}{\langle N\rangle} \ll 1.$$

In fact, its contribution to the density parameter satisfies:

$$\begin{aligned}(\beta\hbar\omega)^3 N_0 &= \rho\lambda_{\mathrm{th}}^3 \frac{N_0}{\langle N\rangle} \\ &\ll \rho\lambda_{\mathrm{th}}^3.\end{aligned}$$

Therefore, the ground-state contribution can be neglected, giving:

$$\boxed{\rho\lambda_{\mathrm{th}}^3 \approx CI(A)}.$$

3.4 (8)

Now N and ω are fixed, while the temperature T is decreased. As long as the ground-state population is negligible, the result of (7) gives:

$$N \approx \frac{C}{(\beta\hbar\omega)^3} I(A) = C\left(\frac{k_B T}{\hbar\omega}\right)^3 I(A), \qquad A = \beta(e_g - \mu) > 0.$$

As T decreases, the prefactor T^3 decreases. To keep N fixed, $I(A)$ must therefore increase. Since $I(A)$ is monotonically decreasing in A, this requires A to decrease towards zero, or equivalently μ to approach e_g.

However, $I(A)$ has the finite maximum $I(0)$. Therefore, the excited states can accommodate at most:

$$N_{\mathrm{ex,max}}(T) = C\left(\frac{k_B T}{\hbar\omega}\right)^3 I(0)$$

particles in the continuous approximation. The critical temperature is determined by:

$$N = C\left(\frac{k_B T_c}{\hbar\omega}\right)^3 I(0).$$

Thus:

$$\boxed{T_c = \frac{\hbar\omega}{k_B}\left(\frac{N}{CI(0)}\right)^{1/3}}.$$

Below T_c, the excited states cannot accommodate all N particles. The remaining particles occupy the ground state, and, in the macroscopic approximation:

$$\boxed{N_0 \approx N - N_{\mathrm{ex,max}}(T) = N\left[1 - \left(\frac{T}{T_c}\right)^3\right]}.$$

3.5 (9)

According to (2), the one-particle energy eigenvalues are:

$$e = e_g + \hbar\omega(n_x + n_y + n_z), \qquad n_x, n_y, n_z = 0, 1, 2, \ldots.$$

Let $s = (e - e_g)/(\hbar\omega)$. The excited states with energy not exceeding e satisfy:

$$n_x + n_y + n_z \le s, \qquad n_x, n_y, n_z \ge 0.$$

For large s, the total number of such states can be approximated by the volume of the corresponding tetrahedron in quantum-number space:

$$\begin{aligned}\mathcal{N}_{\mathrm{ex}}(e) &\approx \int_0^s dn_x \int_0^{s-n_x} dn_y \int_0^{s-n_x-n_y} dn_z \\ &= \frac{s^3}{6} = \frac{1}{6}\left(\frac{e - e_g}{\hbar\omega}\right)^3.\end{aligned}$$

Using the relation suggested in the problem:

$$\mathcal{N}_{\mathrm{ex}}(e) = \int_{e_g}^{e} D(e')\, de',$$

we obtain:

$$\begin{aligned}D(e) &= \frac{d\mathcal{N}_{\mathrm{ex}}}{de} \\ &\approx \frac{1}{2\hbar\omega}\left(\frac{e - e_g}{\hbar\omega}\right)^2.\end{aligned}$$

Comparing this with the assumed form in (6), we find:

$$\boxed{C = \frac{1}{2}}.$$

Equivalently, the same continuous density of states follows from:

$$\begin{aligned}D(e) &\approx \int_0^\infty dn_x \int_0^\infty dn_y \int_0^\infty dn_z\, \delta\big(e - e_g - \hbar\omega(n_x + n_y + n_z)\big) \\ &= \frac{1}{\hbar\omega}\int_0^s dn_x \int_0^{s-n_x} dn_y = \frac{s^2}{2\hbar\omega}.\end{aligned}$$